

FinFlow Loan Management System

Master End-to-End Execution, Integration & Interface Testing Guide

System Overview: FinFlow is an enterprise microservices-based loan application, documentation, and underwriting platform. It connects an Angular frontend with a Spring Boot backend (Auth, Application, Document, Admin services), API Gateway (:8090), Eureka Service Discovery (:8761), Spring Cloud Config (:8888), MySQL relational databases, and RabbitMQ message broker.

1. Exact Build & Execution Command Sequence

To rebuild the Java microservice JAR binaries from source code, clean previous container states, and spin up the complete infrastructure, execute the following commands in exact order from the project root:

```
# Step 1: Navigate to the Root Project Directory
cd "c:\Capg-CoreJAVA-Development\Capgemini-Training\All-Capg-Workspaces\FinFlow Loan Management"

# Step 2: Stop & Remove Existing Containers, Networks and Named Volumes
docker-compose down -v

# Step 3: Rebuild All Java Microservice JARs (Parent POM)
mvn clean package -DskipTests

# Step 4: Build Container Images and Launch Entire Stack
docker-compose up --build -d

# Step 5: Check Running Container Health & Status
docker-compose ps
```

Important Note on Command Flags:

- Correct Spelling: Use `mvn clean package -DskipTests` (capital **T** in `skipTests`, and `clean` instead of `clear`).
- The `-v` flag in `docker-compose down -v` resets MySQL database volumes to a clean state.

2. Master Services, Ports & Credentials Reference

Service / Component	URL / Access Point	Port	Default Credentials / Role
Angular Frontend (Dev)	http://localhost:4200	4200	Applicant: rahul.sharma@example.com / password123
Angular Frontend (Nginx)	http://localhost	80	Admin: admin@finflow.com / password123
Unified Swagger UI	http://localhost:8090/swagger-ui.html	8090	Supports all 4 microservices via Gateway
Eureka Dashboard	http://localhost:8761	8761	N/A (Service Discovery Monitor)
RabbitMQ Console	http://localhost:15672	15672	User: admin Pass: admin
Config Server	http://localhost:8888	8888	Central Git/Native Config

3. Step-by-Step Web Interface User Testing Guide

Test Flow A: User Registration & Sign In

1. Demo Applicant Sign In:

- Open `http://localhost:4200/login`.
- Click the **Quick Demo Login** → **'Applicant Demo'** button.
- Auto-fills Email: `rahul.sharma@example.com` and Password: `password123`.
- Click **Sign In to Dashboard** to access `/applicant/dashboard`.

2. New User Registration:

- Navigate to `/register`.
- **Full Legal Name:** Amit Verma | **Email:** `amit.verma@example.com`
- **Phone Number:** 9876501234 | **Role:** Loan Applicant
- **Password:** SecurePass123 (Observe live green password strength indicator).
- Accept Terms & click **Complete Registration**.

Test Flow B: 6-Step Loan Application Wizard

Navigate to `/applicant/apply-loan` and fill out the fields:

- **Step 1 (Personal):** DOB: 1990-08-20, Address: Flat 501, Blue Ridge, City: Pune, State: Maharashtra, Pincode: 411057.
- **Step 2 (Employment):** Type: Salaried, Employer: TCS Digital, Title: Software Architect, Experience: 8 Years.
- **Step 3 (Financials & DTI):** Monthly Income: **₹1,50,000**, Existing EMI: **₹20,000**, Credit Score: 810.
Notice: DTI ratio auto-calculates to 13.3% with a green 'High Approval Probability' badge.
- **Step 4 (Loan Specs & EMI):** Type: HOME (8.0% rate), Amount: **₹50,00,000**, Tenure: 240 Months.
*Notice: Interactive EMI Calculator updates to **₹41,822/mo** and total interest payable.*
- **Step 5 (Document Upload):** Drag & drop Aadhaar / Salary Slip attachment.
- **Step 6 (Review & Submit):** Check declaration box & click **Submit Loan Application**.

Test Flow C: Admin Underwriting & Decision Desk

1. Sign In as Admin Underwriter:

- Login at `/login` with Email: `admin@finflow.com` / Pass: `password123`.
- Redirection lands on `/admin/dashboard` showcasing portfolio risk charts.

2. Review File & Make Decision:

- Go to `/admin/applications` and click **Review File** on Application #101 or #102.
- Inspect applicant profile, financials, and uploaded documents.
- Enter Remarks: Credit score verified. Approved for disbursement.
- Click **Approve Loan**. Status updates instantly to **APPROVED**, and an audit report event is published to RabbitMQ!

3. System Audit Stream:

- Navigate to `/admin/reports` to inspect live RabbitMQ event logs.

4. Swagger API Sandbox Testing Guide

1. Open **Unified Swagger UI**: <http://localhost:8090/swagger-ui.html>.
2. Use top-right '**Select a definition**' dropdown to choose service (Auth, Application, Document, Admin).
3. **Authenticate**: Call POST `/gateway/auth/login` with `{"email":"john@example.com","password":"password123"}`.
4. Copy the raw JWT token string.
5. Click **Authorize** ■ button, paste token, and click Authorize.
6. **Test Application Service**: Call GET `/gateway/applications/calculate-rate?amount=500000&purpose;=Home Purchase`.
7. **Test Admin Decision**: Call POST `/gateway/admin/applications/101/decision` with payload `{"decisionType":"APPROVED","remarks":"Approved via Swagger"}`.